

# Week 1: Jumping pegs Facilitator guide (W1-F-v1)

This guide is the only place with answers, arguments, timings and teaching notes. The student pages contain problems, boards and recording space and nothing else. Please read your own table's section fully before the session; every answer is worked out here so that nothing has to be solved at the table. Section 9 is background for the two mathematicians.

## 1. The session at a glance

**The move.** A peg jumps over an orthogonally adjacent peg into the empty square just beyond it, and the jumped peg is removed. That is the only move all hour.

**The questions.** All three tables meet the same three questions in different sizes: *Can this pattern of pegs be cleared down to one peg? How far past a line can an army of pegs send a single peg? How few pegs does that take?* The headline facts: on a strip, an army of any size sends a peg exactly one square past the line, never two; on a grid, armies of 2, 4, 8 and 20 pegs reach the first, second, third and fourth row past the line, and no army, however large, reaches the fifth (Conway's Soldiers). The tool that proves the impossibilities is a value attached to each square so that no jump ever raises the total value of the pegs on the board.

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|-----------|-----------------------------------------------------------------------------------------------|
| 0:00–0:05 | Free handling. Pieces and page 1 are on the tables; children do whatever they like with them. |
| 0:05–0:08 | Whole-group launch (section 3): the jump, done by everyone on strip A.                        |
| 0:08–0:50 | Table work (sections 5–7). Roughly forty minutes.                                             |
| 0:50–0:58 | Share (section 8): each table shows one thing.                                                |
| 0:58–1:00 | Pieces back in cups, packets go home.                                                         |

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**Who sits where.** The K–1 section is written so that the parent can run it: it needs reading aloud, scribing, and the answers below, and nothing has to be worked out. Suggested: parent at K–1, second mathematician at grades 2–3, organizer at grades 4–5, with the organizer stopping at the K–1 table twice (around 0:25 for the river strips and around 0:40 for the boards). Swap the mathematicians if you prefer; both sections are self-contained.

## 2. Materials and preparation

- Student packets, printed single-sided (children put pieces on the pages): K–1, 9 pages, 4 copies (3 children + 1 spare); grades 2–3, 9 pages, 5 copies; grades 4–5, 9 pages, 4 copies. Three copies of this guide.
- Pieces. K–1: one checkers set (24 pieces; the last board needs 8 per child) or large counters. The K–1 cells are 1 in square; standard 1-inch checkers fit, 1.25-inch ones overhang a little, which is fine, and pennies work as a backup. Grades 2–3 and 4–5 use pennies (cells are 0.8 in): 10 per child at grades 2–3, 20 per child at grades 4–5 (the row-four board), plus 30 spare. About 130 pennies in all, sorted into a cup per table.
- One pencil per child. A few blank printed grids (any size) at each table for children who want to invent their own boards.
- For the launch: a whiteboard or large sheet on which a strip of five squares can be drawn, and three magnets or large counters; or simply hold up a K–1 page 1 with checkers on it.
- Preparation, about fifteen minutes: print and sort; each adult plays through the first two problems of their table with real pieces; the K–1 adult also plays boards Q, R and T (answers in section 5) so the jumps are familiar.

## 3. The whole-group launch (two to three minutes)

Children stay at their tables; every child has page 1 and pieces. Organizer:

“Find strip A on your first page and put a piece on each gray square. Here is the only move we use today.” (Show it on the board: three squares, pieces on the first two.) “A piece jumps over the piece right next to it and lands on the empty square just beyond. The piece it jumped over comes off. Do it now on your strip.” Wait until every child has jumped and removed a piece; table adults help.

“Three things are not allowed: jumping over an empty square, landing on a square that already has a piece, and jumping slantwise. Today’s question is this: if a bunch of pieces start on one side of a line, how far past the line can they send one piece? Your table has its own way of finding out.” Start the tables. Do not say anything about answers or about the golden ratio; that story belongs to the share.

## 4. Running a table

Every sequence on the pages is *act, record, compare, explain*. Let the children do and record before you ask why. When a child says “it is impossible”, the question to ask, once the recording space is filled, is “How do you know for sure?” Accept explanations a child can produce: pointing at a filled table, listing every case, describing a strategy, showing one counterexample, or saying why the same reason works again on a bigger strip. Do not ask for a rule with a letter in it.

Two habits to enforce gently all hour: the jumped piece really comes off, and pieces are reset to the gray squares before a new try. When a problem asks for “the farthest square” or “the highest row”, the record is the best result over all tries, not the last try.

## 5. Table K–1 (two kindergartners and a first grader; adult reads aloud and scribes)

**Order and timing.** Problems 1–2 (0:08–0:14), 3–4 (0:14–0:24), 5–6 (0:24–0:32), 7–9 (0:32–0:42), 10–12 (0:42–0:50). Read each problem aloud in pieces: one sentence, then the action. Write numbers in boxes for children who ask. Every problem has a real answer and the children should feel that they found it, so hold hints until a child has tried twice.

**Problem 1.** Strip A (pieces on squares 2, 3): the piece on 2 jumps over 3 and lands on 4, or the piece on 3 jumps over 2 and lands on 1; either is a correct jump, and the jumped piece comes off. Strip B (pieces on 3, 4): the leftward jump is 4 over 3, landing on 2.

**Watch for.** Sliding instead of jumping; jumping over an empty square; forgetting to remove the jumped piece. Ask: “Which piece jumped? Which piece came off? Where did it land?”

**Problem 2.** Strip C (pieces on 2 and 4): NO, there is no piece right next to another one. Strip D (all five full): NO, there is no empty square to land on. Strip E (pieces on 2, 3, 4): YES, but only the middle piece can jump (over 3 to 5, or over 2 to 1); each end piece would have to land on an occupied square. Ask which piece can jump and why the end pieces cannot.

**Problem 3.** Strips are six squares long, numbered from the left.

- Strip F (2, 3): always 1 left.
- Strip G (2, 3, 4): always 2 left. Only the middle piece can jump; after it does, the two remaining pieces are far apart and stuck.
- Strip H (2, 3, 5): 1 or 2, and the order matters. Jump 2 over 3 to 4, then 4 over 5 to 6 (or 5 over 4 to 3): one left. But if the first jump is 3 over 2 to 1, the pieces on 1 and 5 are stuck: two left.
- Strip I (1, 2, 4, 5): always 2 left, whatever order.
- Strip J (1, 3, 5): no jump is possible at all; 3 left.
- Strip K (1, 2, 4, 6): 1 or 2. The only first jump is 1 over 2 to 3. Then 3 over 4 to 5 followed by 6 over 5 to 4 leaves one piece; but 4 over 3 to 2 instead leaves pieces on 2 and 6, stuck.

**Watch for.** Children who stop while a jump is still possible. Ask “Is any piece next to another piece with an empty square beyond?”

**Problem 4.** Strips H and K again; the best result is 1 left on both.

**Hints, in order.** “Try starting with the other piece.” “Which jump left two pieces far apart? Can you avoid it?” For strip K after the forced first jump: “Now you have pieces on 3, 4, 6. Two jumps are possible. Try both.”

**Problem 5.** Strips have four home squares (1–4), the river, then square 1 and square 2 across. Strip L (pieces on 3, 4): 3 jumps over 4 and lands on square 1 across; X there. Strip M (2, 3, 4): only the middle piece can move; 3 over 4 lands on square 1 across, or 3 over 2 lands on home square 1; then everything is stuck. Farthest: square 1. Strip N (1, 2, 3, 4): the play is forced. The only possible first jump is 3 over 4 onto square 1 across; the only possible second jump is 1 over 2 to 3; then the pieces on home 3 and on square 1 across are two apart and stuck. Farthest: square 1.

**Watch for.** A child who wants to jump from square 1 across back over home square 4 (it is empty after the crossing, so that is not a jump).

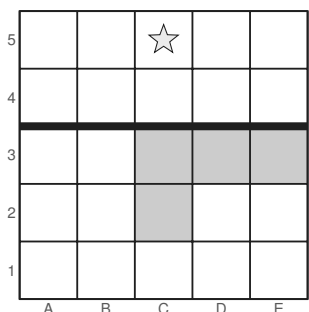
**Problem 6.** Strip O: whatever squares the child chooses, no piece ever reaches the star. Circle 1. (Every choice of home squares was checked by computer; the proof for strips of any length is in section 6, Problem 8, and section 9.) At this table it is enough to compare the strips and to notice, with the child, what a crossing does: the moment a piece lands on square 1 across, home squares 3 and 4 are empty, and to land on the star a piece would need home square 4 full *and* square 1 across full at the same moment. Refilling home square 4 is itself a jump, which uses up two more pieces behind it and leaves two more empty squares. With four pieces in a block the play was forced (Problem 5), so that list of cases is complete; for a placement with gaps, let the child exhaust it.

**Hints, in order.** “What do the two squares just before the river look like right after a crossing?” “What would a piece need in order to land on the star?”

**Problem 7.** Board P: the piece on the lower gray square jumps over the upper one and lands in row 1 across, directly above. X there.

**Problem 8.** Board Q (the two-by-two block): the highest square reached is in row 1 across, never the star. In the natural line each column sends a piece up once, the two crossed pieces then sit side by side in row 1, and they can only jump sideways along that row. Other lines start with a sideways jump inside the block; in every line the highest square reached is in row 1 across. Let the children exhaust it.

**Problem 9.** Board R (the L: two pieces stacked under the star’s column, two more along the river to the right). Three jumps reach the star: the lower stacked piece jumps up over the upper one into row 1; the far right piece jumps left over its neighbor into the square just vacated below the river; that piece jumps up over the row-1 piece onto the star.



Jumps on board R: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5 (the star). (Letters and numbers are for this guide; the K–1 page has none.)

**Hints, in order.** “Which piece will be the last to move, and what must be right under the star just before?” “Get one piece across first. Now the square under it is empty; can something jump into it from the side?”

**Problem 10.** Three pieces never reach the star; the best is row 1 across (two stacked pieces cross, the third cannot help). Complete reason, which a first grader can follow with the pieces: to land on the star you need, at one moment, a piece in row 1 under the star and a piece on the home square directly below it. Making the row-1 piece uses two

pieces and empties the home square below it; the third piece is not on that square (the crossing just emptied it) and cannot jump into it, because a jump needs a second piece to hop over. (Section 6, Problem 14 gives the values version: three home squares are worth at most 18 and the star is worth 21.)

**Problem 11.** Among all ways of putting four pieces on the fifteen home squares of board S, exactly two reach the star: the L of board R and its mirror image (two stacked under the star, two more along the river to the *left*). This was checked by computer for all 1365 placements. Every other four-piece shape stops in row 1 or lower. The block never crosses twice in one column; a row of four along the river cannot cross at all (crossing needs two pieces stacked); a T shape crosses once and is stuck.

**Hints, in order.** “Where were the two stacked pieces in board R? Try putting the other two on the other side.”

**Problem 12.** Board T, eight pieces in a T: five along the river (call them left end, left, middle, right, right end) and three under the middle one (upper, middle, lower). Seven jumps reach the big star in row 3:

1. the upper piece of the column jumps up over the middle river piece into row 1;
2. the lower piece of the column jumps up over the column's middle piece;
3. the left end piece jumps over its neighbor into the middle river square;
4. that middle river piece jumps up over the row-1 piece into row 2;
5. the right end piece jumps over its neighbor into the middle river square;
6. the piece just under it jumps up over it into row 1;
7. the row-1 piece jumps over the row-2 piece onto the big star.

In the letters of the guide (columns A–E, rows 1–7, star at C7): C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. Do it once yourself beforehand. If the children reach only row 2, that is a real result; show the last three jumps and let them redo the whole thing.

**If the table stalls.** Reading is the bottleneck, not the mathematics: shorten by reading only the action sentences. If Problem 3 drags, skip strips I and J and go to the river (Problem 5), which is the heart of the hour. If a child is done listening, hand them a blank grid and ask for “a strip pattern of four pieces that clears to one piece” (for example pieces on squares 1, 2, 4, 6; test any answer by playing it).

**If the table finishes early.** (a) Comb strips: draw a long strip and put pieces on squares 1, 2, 4, 6, 8, 10, . . . ; it can always be cleared to one piece: jump 1 over 2 to 3, so that 3 and 4 form a pair; jump 3 over 4 to 5, so that 5 and 6 form a pair; and so on to the last tooth, where the final pair collapses to one piece. Ask “add one more tooth, does it still work?” (b) A tall column two squares wide: no matter how deep, no piece ever gets past row 1, because with only two columns no sideways jump is possible, so each column is a strip on its own. Let them try a 2-by-4 column. (c) A three-by-three block of nine pieces reaches row 2, but only above a side column of the block, never above its middle column; a three-wide block four deep (twelve pieces) reaches row 3 above a side column. Both facts were checked by computer; let them search.

## 6. Table grades 2–3 (four third graders)

**Order and timing.** Problems 1–2 (0:08–0:12), 3–4 (0:12–0:20), 5–6 (0:20–0:26), 7–9 (0:26–0:34), 10–12 (0:34–0:44), 13–14 (0:44–0:50). Problems 15–16 are for children who finish; most tables will not reach them, and that is fine.

**Problem 1.** Strip A (pegs on 3, 4): 3 over 4 lands on 5, or 4 over 3 lands on 2.

**Problem 2.** Strip B (3 and 5): not possible, no two pegs are adjacent. Strip C (all eight): not possible, no empty landing square. Strip D (3, 4, 5): possible; only the middle peg can jump, landing on 2 or on 6, so X on squares 2 and 6.

**Problem 3.** Eight-square strips, numbered from the left.

- Strip E (2, 3, 5): 1 is possible (2 over 3 to 4, then 4 over 5 to 6 or 5 over 4 to 3); starting with 3 over 2 to 1 strands pegs on 1 and 5, leaving 2. Fewest: 1.
- Strip F (3, 4, 5): always 2.
- Strip G (2, 3, 5, 6): always 2, in every order (there are four first jumps; try them all with the children if they doubt it).
- Strip H (1, 2, 4, 6): 1 is possible: 1 over 2 to 3, 3 over 4 to 5, then 6 over 5 to 4 (or 5 over 6 to 7). Choosing 4 over 3 to 2 as the second jump leaves 2 and 6 stuck. Fewest: 1.
- Strip I (1, 3, 4, 6): always 2.
- Strip J (2–6, five in a row): always 3; the best possible is 3.

A certificate the mathematician can use for F, G and I, after the children have exhausted them: color the squares 1, 2, 3, 1, 2, 3, ... from the left. Every jump changes the number of pegs on *each* of the three colors by one (two colors lose a peg, the third gains one), so the three parities all flip together. A single peg has one odd color and two even ones. Strip F starts with one peg of each color (all odd), strips G and I with two of one color, two of another and none of the third (all even); flipping all three parities never produces one odd and two even, so one peg is unreachable. Strip J has parities one odd, two even and still gets stuck at 3: the certificate can prove impossibility but never possibility.

**Problem 4.** Five-peg patterns that clear to one peg (all of them, on eight squares, with positions from the left): 1-2-3-4-6, 1-2-4-6-8, 1-2-4-7-8, 1-2-5-7-8, 1-3-4-5-6, 1-3-5-7-8, 2-3-4-5-7, 2-4-5-6-7, 3-4-5-6-8, 3-5-6-7-8. Patterns that always end with three or more: five in a row anywhere (3 left), 1-2-3-6-8 (4 left), 1-3-6-7-8 (4 left), and others; any child's pattern is checked by playing it out.

**Problem 5.** Six home squares, the river, then squares 1 and 2 across. Strip M (5, 6): 5 over 6 lands on square 1 across. Strip N (4, 5, 6): only 5 can jump, onto square 1 across (or back to 3); farthest 1. Strip O (3–6): four lines of play, all ending with the farthest square 1. Strip P (1–6): the play is forced: 5 over 6 onto square 1 across, then 3 over 4 to 5, then 1 over 2 to 3; pegs on 3, 5 and square 1 across, all two apart, stuck. Farthest 1.

**Problem 6.** Any placement of up to six pegs: farthest square 1. (Computer check over every subset of the six home squares; section 9 has the proof for strips of any length.)

**Problem 7.** Values on strip S: 1, 1, 2, 3, 5, 8 on the home squares, 13 on square 1, 21 on square 2.

| pegs on                                                | before | after |      |
|--------------------------------------------------------|--------|-------|------|
| all six home squares; the only jump (5 over 8 onto 13) | 20     | 20    | same |
| 2 and 3; left (3 over 2 onto 1)                        | 5      | 1     | down |
| 2 and 3; right (2 over 3 onto 5)                       | 5      | 5     | same |
| 5 and 8; right, across (5 over 8 onto 13)              | 13     | 13    | same |
| 8 and 13; left (13 over 8 onto 5)                      | 21     | 5     | down |

Ask, after the table is filled: “When did the total stay the same? When did it drop? Did it ever go up?” The pattern: a jump to the right (toward the star) removes two neighboring values and lands on the next value, which is exactly their sum, so the total is unchanged; a jump to the left lands on a smaller value than either removed one, so the total drops.

**Problem 8.** Total of the home squares 20; square 2 is worth 21. The argument, in the children's terms: the total starts at 20; no jump ever makes the total bigger (rightward jumps keep it, leftward jumps lower it); if a peg ever stood on square 2 the total would be at least 21; so it never does. Check that every child can say why “at least 21” (the star alone is 21 and all other values are positive). This is a complete proof for strip S, whatever the starting squares (fewer pegs give a smaller total).

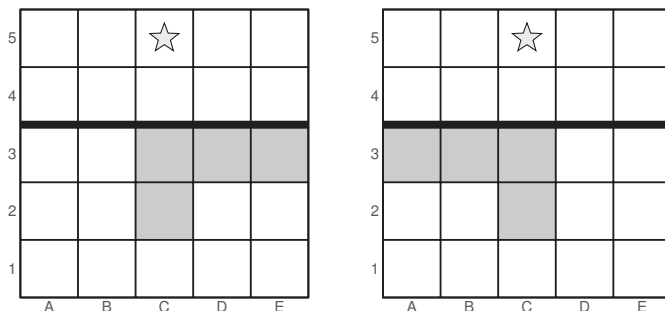
**Problem 9.** Strip T: 1, 1, 2, 3, 5, 8, 13 on the seven home squares, 21 on square 1, 34 on square 2. Home total 33, square 2 is 34. Same reason. The children may notice that the home total is always exactly one less than the star

(12 and 13 on a strip with five home squares; 20 and 21; 33 and 34); this is a property of these numbers (section 9) and worth saying aloud as “the army is always one short”.

**Problem 10.** Board U: C2 over C3 lands on C4.

**Problem 11.** Three pegs never reach C5; the highest row any peg reaches is 4 (for instance from C2, C3 and anything). The reasoning is the same as at K–1, Problem 10, and the values reasoning comes in Problem 14.

**Problem 12.** Only shape 5 works: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5. Shape 1 (block) reaches row 4 only; shape 2 (row of four along the river) cannot cross at all; shapes 3, 4 and 6 reach row 4 only. Among all 1365 ways to place four pegs on the fifteen home squares, only shape 5 and its mirror image reach the star.



**Hints, in order.** “Which square must hold a peg right before the last jump, and which one below it?” “After a peg crosses, refill the square below it from the side.”

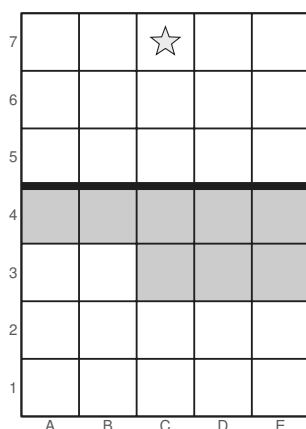
**Problem 13.** Board V values (rows 5 down to 1): 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. (a) C2 over C3 to C4: 5+8 before, 13 after, same. (b) E3 over D3 to C3: 3+5 before, 8 after, same. (c) B3 over C3 to D3: 5+8 before, 5 after, down. (d) C4 over C3 to C2: 13+8 before, 5 after, down. Rule on the board: the value of a square is decided by its distance from the star counted in steps along rows and columns; a jump that ends closer to the star keeps the total, any other jump lowers it.

**Problem 14.** Three most valuable home squares:  $8+5+5 = 18$ . Star: 21. Shape 1 (C3, D3, C2, D2):  $8+5+5+3 = 21$ . Shape 5 (C3, C2, D3, E3):  $8+5+5+3 = 21$ . So three pegs, worth at most 18, can never reach a square worth 21 (the total never rises): this settles Problem 11 for every placement at once. Shape 1 has enough value and still fails: enough value is necessary, not sufficient; the values can only prove that something is impossible, never that it is possible. Shape 5 spends its 21 exactly: each of its three jumps keeps the total, and the lone peg on the star is worth 21.

**Problem 15.** Board W (T of eight): C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. Values (section 7, Problem 11): the eight squares are worth exactly 55, the star’s value, so every jump in a solution must keep the total and the solution ends with a single peg.

**Hints, in order.** “What must be on C6 and C5 just before the end?” “Get a peg to C5 first, then refill C4 from the side, then jump C4 over C5.” “Now C5 is empty again and C6 is full: you need another peg on C5 with a peg on C4 under it.”

**Problem 16.** The only other eight-peg sets on board X that reach C7 are the two “flag” shapes: A4 B4 C4 D4 E4 with C3 D3 E3, and its mirror image A4 B4 C4 D4 E4 with A3 B3 C3 (checked by computer: any eight-square set worth at least 55 must contain C4, B4, D4, C3, and then either four of the 5-valued squares, none of which works, or three of them and one of the five 3-valued home squares of the board, giving 50 sets of which exactly the T and these two work).



Right flag: E3 over E4 to E5; D3 over D4 to D5; E5 over D5 to C5; C4 over C5 to C6; A4 over B4 to C4; C3 over C4 to C5; C5 over C6 to C7.

Left flag (mirror image): C3 over C4 to C5; E4 over D4 to C4; C4 over C5 to C6; B3 over B4 to B5; A3 over A4 to A5; A5 over B5 to C5; C5 over C6 to C7.

**If the table stalls.** Problem 3 can absorb a whole hour; after ten minutes move on even if strips I and J are unfinished. If the values table (Problem 7) meets resistance, do the rows together with pennies on the strip, saying each sum aloud (“5 and 8 go away, 13 arrives”). If Problem 15 is too long, give jumps 1–3 and let them find 4–7.

**If the table finishes early.** (a) Ask for the values on a strip with nine home squares and whether the army is again one short (1, 1, 2, 3, 5, 8, 13, 21, 34 sum to 88; square 2 is worth 89). (b) Give board T from the grades 4–5 packet (twenty pegs, star in row 4; solution in section 7, Problem 13). (c) Blocks: a two-wide column of any depth never passes row 1 (each column is a strip; no sideways jump exists); a three-by-three block reaches row 2 above a side column; five pegs along the river with five more under them (five wide, two deep) reach row 3; a five-by-four block of twenty pegs reaches only row 3. All checked by computer.

## 7. Table grades 4–5 (two fourth graders and a fifth grader)

**Order and timing.** Problems 1–2 (0:08–0:13), 3–4 (0:13–0:22), 5–7 (0:22–0:31), 8–9 (0:31–0:36), 10–11 (0:36–0:45), 12 (0:45–0:50). Problem 13 (row four, nineteen jumps) is for a group that finishes; it is a genuine puzzle and the answer is below.

**Problem 1.** Strip A (pegs on 5, 6 of the six home squares): 5 over 6 lands on square 1 across.

**Problem 2.** Strip B (4, 5, 6): only 5 can jump; farthest square 1. Strip C (3–6): four lines of play, each ending with the farthest square 1. Strip D (1–6): forced play, 5 over 6 across, 3 over 4 to 5, 1 over 2 to 3, stuck; farthest 1. Strip E: whatever the placement, 1. (Every subset of the six home squares was checked by computer; the proof for any strip is Problems 3–4 and section 9.)

**Problem 3.** Values 1, 1, 2, 3, 5, 8 on the home squares, 13 on square 1, 21 on square 2.

| pegs on                                                | before | after |      |
|--------------------------------------------------------|--------|-------|------|
| all six home squares; the only jump (5 over 8 onto 13) | 20     | 20    | same |
| 2 and 3; left (3 over 2 onto 1)                        | 5      | 1     | down |
| 2 and 3; right (2 over 3 onto 5)                       | 5      | 5     | same |
| 5 and 8; right, across (5 over 8 onto 13)              | 13     | 13    | same |
| the second 1 and 2; left (2 over 1 onto the first 1)   | 3      | 1     | down |
| 8 and 13; left (13 over 8 onto 5)                      | 21     | 5     | down |

After the table: a jump toward the star removes two neighboring values and lands on the next one, which is their sum (the values are built that way), so the total is unchanged; a jump away from the star lands on a value smaller than either removed value, so the total drops. The total never goes up.

**Problem 4.** Home total 20, square 2 worth 21. Proof for strip F: the total starts at 20 (or less, with fewer pegs); no jump raises it; a peg on square 2 would make it at least 21. Strip G: 1, 1, 2, 3, 5, 8, 13 / 21 / 34; home total 33,

square 2 worth 34; the same argument word for word. Ask “Why is the army always exactly one short?” and let them test  $1+1+2+3+5 = 12$  against 13 and  $1+1+2+3+5+8+13+21 = 54$  against 55: adding each Fibonacci number to the sum of all earlier ones gives the next one minus 1 (section 9). Ask also why the same reason works for a strip of any length: the values can be continued as far as needed, and the army is one short every time.

**Problem 5.** Board O: C2 over C3 lands on C4.

**Problem 6.** Three pegs never reach C5; the highest row reached is 4 (e.g. from C2, C3 and any third square). Full reason: see Problem 9.

**Problem 7.** Only shape M: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5. Shape I (block) reaches row 4; shape J (row along the river) never crosses; shapes K, L, N reach row 4. Among all 1365 placements of four pegs on the fifteen home squares, only shape M and its mirror image reach the star (computer check).

**Hints, in order.** “Which two squares must both be full just before the last jump?” “Once a peg has crossed, the square under it is empty; refill it from the side.”

**Problem 8.** Board P values (rows 5 to 1): 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. (a) C2 over C3 to C4: 13 before, 13 after. (b) E3 over D3 to C3: 8 and 8. (c) B3 over C3 to D3: 13 before, 5 after. (d) C4 over C3 to C2: 21 before, 5 after. The rule: a square’s value depends on its distance from the star in steps along rows and columns (the star is 21, its four neighbors 13, and so on down the Fibonacci numbers); a jump that ends nearer the star keeps the total, any other jump lowers it.

**Problem 9.** Best three home squares  $8+5+5 = 18$ ; star 21; shape I  $8+5+5+3 = 21$ ; shape M  $8+5+5+3 = 21$ . So three pegs, placed anywhere, are worth at most 18 and can never reach a square worth 21: Problem 6 is settled for all placements at once. Shape I is worth 21 and still fails, so enough value does not guarantee success; the values only ever prove impossibility. Shape M spends its 21 exactly, each jump keeping the total, and ends with one peg on the star.

**Problem 10.** Board Q: C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. A group that plans backward (“C7 needs C6 and C5 full; C6 needs C5 and C4 full ...”) finds it in a few minutes; a group jumping at random may need the first hint.

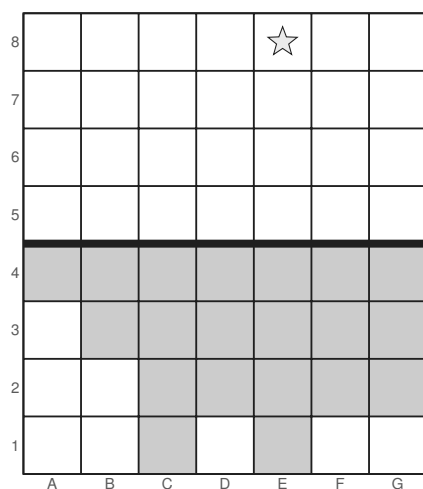
**Hints, in order.** “Plan the last jump first.” “The column of three under the middle is there to fill C5 twice.” “The two ends of the row of five are there to fill C4 twice.”

**Problem 11.** Board R values (rows 7 to 1): 21 34 55 34 21 / 13 21 34 21 13 / 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. Seven most valuable home squares:  $13+8+8+8+5+5+5 = 52$ ; star 55; the T of board Q:  $13+8+8+5+5+8+5+3 = 55$ . So seven pegs never reach C7 (52 is less than 55, and the total never rises), and any eight-peg army that does must be worth at least 55 and spend it exactly.

**Problem 12.** The only other eight-peg sets that reach C7 are the two flags: A4 B4 C4 D4 E4 with C3 D3 E3 (E3 over E4 to E5; D3 over D4 to D5; E5 over D5 to C5; C4 over C5 to C6; A4 over B4 to C4; C3 over C4 to C5; C5 over C6 to C7) and its mirror image A4 B4 C4 D4 E4 with A3 B3 C3 (C3 over C4 to C5; E4 over D4 to C4; C4 over C5 to C6; B3 over B4 to B5; A3 over A4 to A5; A5 over B5 to C5; C5 over C6 to C7). Why nothing else: an eight-square set worth at least 55 must contain C4 (13) and B4, D4, C3 (8 each), and then either four of the five 5-valued squares (57 in all; none of the five such sets works) or three 5-valued squares and one of the five 3-valued home squares of the board (55 exactly; of these 50 sets, exactly the T and the two flags work). Both statements were checked by exhaustive computer search.

**Problem 13.** Board T (columns A–G, rows 1–8, river above row 4, star at E8). Twenty pegs: A4 B4 C4 D4 E4 F4 G4 (row 4); B3 C3 D3 E3 F3 G3 (row 3); C2 D2 E2 F2 G2 (row 2); C1 E1 (row 1). Nineteen jumps:





1. G3 over G4 to G5
2. F3 over F4 to F5
3. G5 over F5 to E5
4. E4 over E5 to E6
5. E2 over E3 to E4
6. G2 over F2 to E2
7. E1 over E2 to E3
8. E3 over E4 to E5
9. E5 over E6 to E7
10. D3 over D4 to D5
11. C3 over C4 to C5
12. C5 over D5 to E5
13. C1 over C2 to C3
14. B3 over C3 to D3
15. D2 over D3 to D4
16. A4 over B4 to C4
17. C4 over D4 to E4
18. E4 over E5 to E6
19. E6 over E7 to E8

The structure is worth pointing out: jumps 1–9 use only the ten pegs in columns E–G (a three-by-three block plus E1) and park a peg on E7; jumps 10–12 bring a peg from the left half to E5; jumps 13–17 bring a second peg to E4 while E5 is full; jumps 18–19 finish. With Fibonacci values scaled so that the star is 377, the twenty squares are worth exactly 377, so every jump must keep the total. Nineteen pegs can be worth 377 too (the sixteen squares within seven steps of the star plus three of the nine squares at eight steps), but none of those 84 sets works, by exhaustive computer search; the literature (section 9) also gives twenty as the minimum.

**Hints, in order.** “The last jump is E6 over E7 to E8, so plan to park a peg on E7 and then bring another to E6 with E5 full.” “The right three columns and the column under E are a row-three army; use them first.” “After the peg is on E7, the left half must deliver a peg to E4 while E5 is full.”

**If the table stalls.** Problems 3–4 are the heart; give them time and let the children do the sums with pennies on the strip. If Problem 10 goes past ten minutes, give jumps 1–3. Board T is optional.

**If the table finishes early.** (a) Row five: tell them the true story (section 8) and ask what values would be needed, given that the argument must work for an army of any size; they will see that Fibonacci numbers run out of “room” on an infinite board and the mathematician can introduce the number  $\sigma$  with  $\sigma^2 + \sigma = 1$ . (b) Narrow armies: a two-wide column of any depth never passes row 1 (no sideways jump is possible, so each column is a strip); a three-wide army of any depth never reaches row 4 in any column (values: with the target above the middle column the three columns are worth  $\sigma^2 + 2\sigma^3 = 3\sigma - 1 \approx 0.85 < 1$  in the units of section 9, and less above any other column), though a three-by-four block does reach row 3, above a side column. (c) Blocks: a five-by-four block of twenty pegs reaches only row 3, a five-by-five block reaches row 4 (computer checks); ask them to find a twenty-peg shape of their own that reaches row 4.

## 8. The share (eight minutes)

Order: K–1, grades 2–3, grades 4–5, then the organizer. Two minutes each; children hold up pages and pieces, adults keep it moving.

- K–1 shows strip N (four pieces, the play is forced, the farthest square is 1) and board R (the L shape reaching the star in three jumps).
- Grades 2–3 show strip S with the values: the six pegs are worth 20, square 2 is worth 21, and no jump ever makes the total bigger. “The army is always one short.”
- Grades 4–5 show the T of eight reaching row 3, and, if they got there, board T with twenty pegs reaching row 4.
- Organizer, one minute: on an endless grid with an endless army below the line, John Conway asked how high a single peg can be sent. Four rows, with twenty pegs, as just shown; and never five, not with any army, however large, in any number of moves. The proof is the same idea the tables used, values that a jump can never increase, with one special number in place of the Fibonacci numbers. Next time we can look at that number.

## 9. The mathematics behind the session (for the two mathematicians)

**The potential.** Fix the target square  $T$  and give the square at Manhattan distance  $d$  from  $T$  the weight  $\sigma^d$ , where  $\sigma = (\sqrt{5} - 1)/2 \approx 0.618$  satisfies  $\sigma^2 + \sigma = 1$ . Consider a jump along three collinear squares at distances  $d_1, d_2, d_3$  from  $T$  (jumper, jumped, landing). Either the jump moves toward  $T$  along that line,  $(d+2, d+1, d)$ , and the weight change is  $\sigma^d - \sigma^{d+1} - \sigma^{d+2} = \sigma^d(1 - \sigma - \sigma^2) = 0$ ; or it moves away,  $(d, d+1, d+2)$ , change  $\sigma^{d+2} - \sigma^d - \sigma^{d+1} < 0$ ; or it crosses the line through  $T$ ,  $(d+1, d, d+1)$ , change  $-\sigma^d < 0$ . So the total weight  $W$  of the pegs on the board never increases:  $W$  is a monovariant. Reaching  $T$  needs  $W \geq 1$  at that moment, so an initial position with  $W < 1$  can never reach  $T$ . This is the argument of Berlekamp, Conway and Guy (*Winning Ways*, vol. 4, ch. 23, “Purging Pegs Properly”) and of Aigner’s “Moving into the desert with Fibonacci” (Math. Magazine 70, 1997); Honsberger’s *Mathematical Gems II* also has a chapter on it.

**Row five is unreachable.** With  $T$  in row 5, the column under  $T$  has weight  $\sigma^5 + \sigma^6 + \dots = \sigma^5/(1 - \sigma) = \sigma^5/\sigma^2 = \sigma^3$ , and the columns at horizontal distance  $k \geq 1$  have weight  $\sigma^{3+k}$  each. The whole half-plane is worth  $\sigma^3 + 2 \sum_{k \geq 1} \sigma^{3+k} = \sigma^3 + 2\sigma^4/\sigma^2 = \sigma^3 + 2\sigma^2$ . Using  $\sigma^2 = 1 - \sigma$  and  $\sigma^3 = \sigma - \sigma^2 = 2\sigma - 1$ , this is  $2\sigma - 1 + 2 - 2\sigma = 1$ : the infinite army is worth exactly the target square. A finite army is worth strictly less than 1, and  $W$  never increases, so no finite army reaches row five; and any actual play uses finitely many pegs. (For  $T$  in row 4 the half-plane is worth  $1/\sigma = \phi > 1$ , which is why the argument does not exclude row 4, and indeed 20 pegs suffice.)

**One dimension.** On a strip with  $T$  two squares past the line, the home squares are worth  $\sigma^2 + \sigma^3 + \dots = 1$  exactly, and one square past the line the whole home side is worth  $\sigma^{-1} > 1$ . So a finite army reaches square 1 (two pegs do) and never square 2. Whether the army is contiguous or has gaps makes no difference. The Fibonacci version used on the pages is the same argument with the values  $F_1, F_2, \dots$  read from the far end of a finite strip: a jump toward  $T$  replaces  $F_k + F_{k+1}$  by  $F_{k+2}$ , and  $\sum_{i=1}^n F_i = F_{n+2} - 1$ , so a home side of  $n$  squares is worth  $F_{n+2} - 1$  while square 2 is worth  $F_{n+2}$ : “always one short”. The identity is proved by induction (adding  $F_{n+1}$  to both sides turns the case  $n$  into the case  $n+1$ ), and the children can discover it as the pattern 12/13, 20/21, 33/34, 54/55. Integer values work on any finite board because all values are positive there; on an infinite board the integer recurrence would have to run backward into 0, 1,  $-1, \dots$ , and the geometric  $\sigma^d$  is the right replacement.

**Two dimensions on the pages.** The value boards use  $F_{n-d}$  at distance  $d$  with  $n$  large enough that every printed square has a positive value (board P: star 21; board R: star 55; for board T take the star to be 377). Everything the pages ask is then exact arithmetic: three home squares are worth at most  $8 + 5 + 5 = 18 < 21$ , seven at most  $13 + 8 + 8 + 8 + 5 + 5 + 5 = 52 < 55$ , so 3 and 7 pegs fail for rows 2 and 3, and 4 and 8 succeed by exhibition. The minimal armies for rows 1–4 are 2, 4, 8, 20 (*Winning Ways*; Bell, Hirschberg and Guerrero-García, “The minimum size required of a solitaire army”, INTEGERS 7 (2007), treat this and related boards). For row 4 the potential gives only 19 as a lower bound: the nineteen heaviest home squares (the sixteen within seven steps and three at eight steps) are worth exactly 1, i.e. 377 on board T. A nineteen-peg solution would therefore have to be *perfect*, every jump toward the target and the final position a single peg; an exhaustive search over the 84 candidate sets, restricted to weight-preserving jumps, finds none, which is how “20” was confirmed for this guide. The twenty-peg set on board T is also worth exactly 377, so its nineteen jumps are all weight-preserving, and it ends with one peg. The exact-equality cases are the subtle part of the whole subject: for row 3, exactly three eight-peg sets work (the T and the two flags), and for row 2 exactly the two L shapes.

**Sufficiency is a different matter.** The block of four (worth 21) and the T of four (worth 23) cannot reach row 2; the value argument is a certificate of impossibility only. Deciding reachability in general is hard: solvability of generalized peg solitaire is NP-complete (Uehara and Iwata, 1990), which is why the positive answers on the pages come from explicit jump sequences found by search.

**Pagoda functions.** In peg solitaire proper (clear a board to one peg), a function  $p$  on the squares with  $p(a) + p(b) \geq p(c)$  for every jump  $a$  over  $b$  to  $c$  is called a pagoda function; the total of  $p$  over the pegs never increases, so a target position worth more than the start is unreachable. Our value boards are pagoda functions for the solitaire-army problem, and the mod-3 coloring used at the grades 2–3 table (three colors, every jump flips the parity of all three

counts) is the classical parity certificate of *Winning Ways*; both are necessary conditions only, as strip J and shape I show. Beeler and Hoilman ("Peg solitaire on graphs", Discrete Math. 311, 2011) start the study of the same game on arbitrary graphs.

**Facts checked by computer for this session** (any of them can be handed to a fast table): all 1365 four-peg placements on the fifteen home squares of the row-2 board were searched, and only the two L shapes reach the star; all eight-peg sets of sufficient value on the row-3 board were searched (T and two flags); the 1-D strips of up to eight home squares were searched over every subset; rectangular blocks of pegs with their top row on the line reach at most the following rows, in the best column: two wide (any depth tried, and provably any depth) row 1; three by two, three by three, four by two, four by three: row 2; three by four, four by four, five by two, five by three, five by four: row 3; five by five: row 4. A three-wide army of any depth cannot reach row 4 in any column, by the potential: with the target above its middle column its three columns are worth  $\sigma^2 + 2\sigma^3 = 3\sigma - 1 \approx 0.854$ , above a side column  $\sigma^2 + \sigma^3 + \sigma^4 = 2 - 2\sigma \approx 0.764$ , and above any other column less still.

**Where it leads.** The monovariant is the basic tool of olympiad combinatorics (invariants and semi-invariants), and the same idea appears as potential functions in amortized analysis of algorithms and as Lyapunov functions in dynamics: to show a process cannot reach a state, find a quantity that only moves one way. The golden ratio enters because the jump rule forces the recurrence  $v_{k+2} = v_k + v_{k+1}$ ; the Fibonacci strip values, Zeckendorf representations and the continued fraction of  $\phi$  all sit next door. For a next session: the solitaire army on the triangular or hexagonal lattice and in three dimensions (the same potential with a different sum gives different maximal rows), peg solitaire on small graphs (which trees can be solved?), and the reverse question of reconfiguration: which positions can reach which.